

Apartment EV Parking Shared Charging Case Study

ODD+D case brief, expected action situations, and current research-design gaps

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Current status: the case is conceptually strong enough for pilot AS-extraction tests, but it still needs empirical reference values, formal queue-policy variants, validation criteria, and a clearer experimental design before it should be treated as a complete ABM study.

1. Case Definition

This original case studies shared electric-vehicle charging in an apartment parking facility. Residents receive a discounted per-kWh charging rate, non-resident users pay the regular per-kWh rate, and all users compete for access to a small number of shared charging bays. The research focus is fair queue access rather than tariff optimization.

The model setting is a mixed resident and non-resident garage where EV adoption has grown faster than charger capacity. The resident discount changes demand and perceived entitlement, but it does not formally grant queue priority. The central governance problem is how scarce charging access is ordered, enforced, and released over time.

2. What the Case Already Has

- A stable ODD+D description covering purpose, entities, state variables, spatial and temporal scales, learning, sensing, interaction, collectives, heterogeneity, stochasticity, observation, implementation, initialization, input data, and submodels.
- A model-facing game scenario that explains mechanisms without listing or numbering the expected AS labels.
- A hidden evaluation reference with six expected action situations for manual scoring and future automated evaluation.
- A spreadsheet version aligned with the electricity ODD template, with a separate Expected AS sheet so the gold answers are visible but not mixed into the model input.

3. Main Gaps to Fill

Gap area	What is missing now	How to complete it
Empirical grounding	The ODD lists possible data sources, but no actual reference values are defined for charger count, resident EV ownership, arrival rates, discount size, fault rates, or overstay frequency.	Create a parameter table with baseline, low, and high values. Mark which values come from logs, surveys, literature, or assumptions.
Queue rule specification	The case says queue fairness is central, but the exact rule is not fully operationalized.	Define FIFO, reservation windows, no-show cancellation, overstay grace period, resident and non-resident eligibility, and how staff override rules.
Fairness definition	Fair queue access is the core concept, but the normative metric is still broad.	Choose measurable criteria such as wait-time inequality, access share by group, queue-order violations, complaint rate, and completed-session reliability.
Decision rules	The agent logic is described at a high level, but thresholds and update rules are not yet precise.	Write thresholds for reserving, joining, bypassing, complaining, moving out, and enforcing; define how experience changes probabilities.
Policy experiments	The case identifies governance mechanisms but does not yet define experiment arms.	Compare baseline FIFO, resident quota, idle-time penalty, stricter enforcement, reservation caps, charger expansion, and maintenance-priority policies.
Validation plan	There is no current test for whether simulated waiting, utilization, complaints, or violations match plausible real-world behavior.	Use face validation, sensitivity analysis, extreme-condition tests, and comparison with observed session or complaint logs when available.
Privacy and ethics	The case uses residency status, charger logs, complaints, and possible monitoring, but privacy limits are not specified.	Define what data are anonymized, who can inspect logs, how staff discretion is audited, and how non-resident users are treated fairly.

4. Expected Action Situations

These six ASs should be treated as the current gold set for evaluation. They should not be included in prompts used to test whether a model can extract ASs from the ODD and scenario text.

AS	Expected action situation	Evaluation meaning
AS1	Queue-order and prompt move-out compliance between EV users	Captures whether users respect the posted queue and release scarce charging bays promptly.
AS2	Sequential learning in reservation and off-peak charging behaviour	Captures how visible successful or opportunistic behavior changes later user choices.

AS	Expected action situation	Evaluation meaning
AS3	Shared charger-capacity contribution dilemma among residents	Captures the resident-level dilemma around paying for expanded capacity or improved monitoring.
AS4	Informal priority exchange between EV user and parking staff	Captures staff discretion and informal access that can weaken queue legitimacy.
AS5	Formal complaint and enforcement coordination between user and management	Captures whether users report problems and whether management responds with enforcement or repair.
AS6	Shared charger-occupancy common-pool congestion between EV users	Captures congestion caused by long or unnecessary charger-bay occupation under per-kWh-only pricing.

5. Recommended Next Steps

1. Decide the exact fairness definition before adding more scenario text. The strongest candidates are queue-order compliance, wait-time inequality, and charger-access reliability by user category.
2. Create a parameter table with baseline values and ranges. This will make the ABM reproducible and easier to defend.
3. Write policy experiment arms before coding the model. At minimum, compare baseline FIFO, idle-time penalty, stricter enforcement, resident quota, and charger expansion.
4. Keep the gold AS list outside the model prompt. Use it only for scoring TP, FN, FP, precision, and recall.
5. After the case text is finalized, run the same 30-run ODD-only and ODD+game extraction protocol used for water and electricity.

6. Prompting and Evaluation Boundary

For model testing, the input should include the ODD text and, when testing ODD+game, the game scenario context. The Expected AS sheet and scenarios reference should remain hidden from the model. Evaluation should count only true positives, false negatives, false positives, precision, and recall against the six-AS gold set.